

Atharva Bapat

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SUMMARY

Engineer with production experience building end-to-end LLM applications, RAG pipelines, and agentic AI systems on AWS. Proven track record delivering measurable business impact through prompt engineering, evaluation frameworks, and CI/CD-integrated ML workflows. Masters in Computer Science graduate from UNC Charlotte.

SKILLS & INTERESTS

Skills: Agentic AI, Retrieval-Augmented Generation (RAG), Statistical Analysis, Time-Series Data, AWS Serverless Services, Model Monitoring / MLOps, Vector Embeddings, Machine Learning, Large Language Models (LLMs), Prompt Engineering, API development, FastAPI, Experimental Design, Anomaly Detection / Forecasting, Natural Language Processing (NLP), Data Visualization, CI/CD (GitLab), Hallucination Reduction

EDUCATION

University of North Carolina at Charlotte

Charlotte, NC

Master's Degree in Computer Science

Graduation Date: May 2024

WORK EXPERIENCE

Theoremlabs Partners LLC

Charlotte, NC

Programmer Analyst

Sep 2024 - Present

- Engineered a production **RAG pipeline** combining vector embeddings and LLMs for domain-specific financial Q&A, reducing advisor research time by 60%; implemented A/B testing and evaluation frameworks to reduce hallucinations and improve retrieval accuracy.
- Architected real-time data integration pipelines consuming RabbitMQ message queues to synchronize financial data across systems, ensuring analytical accuracy for downstream reporting and AI model inputs.
- Partnered with financial advisors and business teams to translate complex requirements into AI-driven automation workflows, reducing manual research overhead and expanding access to domain-specific knowledge through **LLMs**.
- Deployed scalable, serverless cloud architectures on **AWS** (Lambda, Step Functions, API Gateway, S3, SQS) with automated **CI/CD** pipelines via GitLab, enabling rapid release of AI microservices and reducing time-to-production.
- Implemented structured reasoning traces and confidence scoring in LLM outputs to support explainability, enabling financial advisors to audit and trust AI-generated recommendations.

Persistent Systems

Pune, India

Software Developer

Jan 2022 - Jun 2022

- Collaborated with cross-functional teams to translate business requirements into scalable data solutions, enabled dynamic infrastructure observability by building **React + TypeScript** dashboards integrated with AWS APIs for real-time.
- Ensured secure and flexible data access by implementing AWS API Gateway over Lambda APIs, allowing users to retrieve data by custom time ranges while preserving privacy.

PROJECT EXPERIENCE

LLM-Based Natural Language Querying for Databases

- Designed a structured knowledge layer using MCP protocol to connect domain context, entity relationships, and database schemas — enabling knowledge-grounded natural language querying.
- Enabled intuitive self-service analytics, reducing dependency on analytics teams and accelerating business decision-making.
- Applied **prompting strategies** to evaluate prompt effectiveness and reducing hallucinations by the model.

Agentic AI Voice Platform — Autonomous Industrial

- Designed human-in-the-loop fallback logic and agentic decision flows enabling the system to reason across multi-step tasks with minimal human intervention and **projected to reduce operational costs by 30%** for client organizations.
- Integrated real-time web scraping pipelines to feed the agent live, domain-specific intelligence for **context-aware decision-making** during active calls.
- Designed core **conversation flows** and **automation workflows**, defining how the agent reasons, responds, and executes multi-step tasks across diverse industry use cases.

DocumentIQ: Intelligent Financial Document Process

- Built an **AI Automation** data integration pipeline using Hugging Face LayoutLM for structured data extraction from scanned **financial and government documents (KYC, tax forms)**.
- Designed and implemented an interactive Streamlit interface with a **FastAPI backend** for **OCR processing**, validation, and database integration.
- Reduced manual data entry by **80%** by automating structured data extraction from KYC and tax documents using LayoutLM.